

Stephen Phillips

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EDUCATION

University of Pennsylvania

Ph.D. in Computer Science (Advisor: Kostas Daniilidis)

August 2014 - May 2021

University of Pennsylvania

M.Sc. in Computer Science

August 2014 - June 2016

University of California, Los Angeles

B.S. in Computer Science (GPA 3.97)

August 2010 - June 2014

TEACHING EXPERIENCE

Engineering 27/Computer Science 72 Computer Vision

January 2022 - May 2023, two times

Instructor

Swarthmore College

- Sophomore/junior level, 15 students
- Topics: Fundamentals of computer vision, image processing, neural networks, and 3D reconstruction.
- Updated 20 lectures, 12 homework assignments, and four projects. Fully designed two lectures and an open-ended capstone project.

Engineering 71 - Digital Signal Processing

January 2022 - May 2022

Instructor

Swarthmore College

- Junior/senior level, five students
- Topics: Discrete time signals, Z-transforms, Fourier Transforms, and the Fast Fourier Transform.
- Fully designed all lectures and projects, with homework from textbook. Designed projects on DNA correlation, linear predictive coding, and sound classification with band-pass filters.

Engineering 15 - Fundamentals of Digital and Embedded Systems

August 2021 - December 2022, two times

Co-instructed with lab instructor Professor Erik Cheever

Swarthmore College

- Freshman level, 50 students
- Topics: Introducing programming with Arduino, and laboratory assignments on hardware implementations.
- Designed lectures, homework assignments, and exams. Homework used Adafruit Circuit Playground, implemented in Adafruit's C language variant.

Engineering 19 - Numerical Methods and Applications in Engineering

August 2021 - December 2021

Instructor

Swarthmore College

- Sophomore level, 19 students
- Topics: Introduction to Python, NumPy, and SciPy, basic numerical method algorithms and analysis.
- Designed all 22 lectures, 12 homework assignments, two exams, and five projects, and one open-ended final project.

CIS 107/VLST 209 - Visual Culture through the Computer's Eye

January 2021 - June 2021

Co-instructor and co-designer with Dr. Will Schmenner.

University of Pennsylvania

- Junior level, online, 15 students (approximately evenly split between Humanities and Engineering)
- Topics: Intersection of computer vision and visual studies, dataset and algorithm design, discussion and writing on the real-world stakes of computer vision and the digital humanities.
- Designed engineering-oriented lectures, facilitated class discussions, mentored student teams in designing datasets.

ACADEMIC EXPERIENCE

Visiting Assistant Professor

September 2021 - May 2023

Swarthmore College

Swarthmore, PA

- Teaching focused two year professorship
- Mentored two students on senior thesis projects. See Teaching section for classes taught.
- Mentored five undergraduate students in extracurricular multi-sensor fusion and machine learning projects, all five continued to graduate programs in CS/engineering.

TEACHING ASSISTANT EXPERIENCE

MCIT 515 - Linear Algebra for Machine Learning

Head teaching assistant under Professor Jean Gallier

July 2020 - December 2020

University of Pennsylvania

- Master's level, online, 25-35 students. Students are not expected to have any background in Computer Science.
- Topics: Basics of linear algebra to prepare students for machine learning applications. Link to MCIT program page.
- Graded homework assignments, developed a project auto-grader, and held office hours.

MEAM620 - Advanced Robotics

Lecturer and teaching assistant, under various instructors

January 2016 - May 2019, three times

University of Pennsylvania

- Master's level, 30-40 students
- Topics: Robotic path planning, control theory, and machine perception/computer vision for quadrotors, with projects using real quadrotors. Link to course wiki.
- Held office hours, designed an auto-grader, assisted with hardware testing for student projects, and designed and delivered six lectures on computer vision for robotics

CIS390 - Robotics: Planning and Perception

Teaching assistant/course designer under Prof. Daniilidis

September 2015 - December 2015

University of Pennsylvania

- Sophomore level, 20 students
- Topics: New course in robotic path planning algorithms with Python projects.
- Course designed from scratch with other teaching assistants; primary responsibility was designing a simulation environment to test robotic planning algorithms for course coding projects.

Coursera - Robotics: Perception

Teaching assistant and homework designer

January 2016 - July 2016

University of Pennsylvania (Online)

- General audience, online, (large-scale MOOC; 20-40 students active)
- Topics: Perception module of the UPenn Robotics Specialization on Coursera, focused on basics of computer vision such as the image formation process up to multi-view geometry. Link to course page.
- Designed four homework assignments, wrote supporting teaching material, and wrote an auto-grader for all assignments.

PUBLICATIONS

Xiaoyi Cai, Siddharth Ancha, Lakshay Sharma, Philip R Osteen, Bernadette Bucher, **Stephen Phillips**, Jiuguang Wang, Michael Everett, Nicholas Roy, and Jonathan P How. 2024. Evora: Deep evidential traversability learning for risk-aware off-road autonomy. *IEEE Transactions on Robotics* (2024).

Stephen Phillips and Kostas Daniilidis. 2019. All graphs lead to Rome: Learning geometric and cycle-consistent representations with graph convolutional networks. *CVPR 2019 Image Matching: Local Features and Beyond Workshop* (2019).

Andrew Jaegle, **Stephen Phillips**, Daphne Ippolito, and Kostas Daniilidis. 2018. Unsupervised learning of image motion by recomposing sequences. *International Conference on Learning Representations (ICLR)* (2018).

Andrew Jaegle, **Stephen Phillips**, and Kostas Daniilidis. 2016. Fast, robust, continuous monocular egomotion computation. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA) 2016*, 773–780.

INDUSTRY EXPERIENCE

Robotist, Applied Scientist

Robotics and AI Institute

May 2023 - Present

Cambridge, MA

- Designed a hardware test workflow for team's trained reinforcement-learning based policies for safe deployment on physical robots outside of simulation.
- Developed a sonar-camera fusion pipeline for indoor navigation on our legged robot platform, reducing collisions in low-texture and cluttered environments.
- Built semantic mapping pipelines combining self-supervised visual features (e.g., DINO) with LiDAR odometry to enable semantic-based terrain traversability assessment downstream.

Waymo Perception Research Intern

May 2021 - July 2021

*Waymo**Remote*

- Conducted research on multi-sensor fusion of camera and radar models for autonomous driving in adverse weather conditions.
- Implemented camera-radar sensor data fusion components for the model training pipeline.

Google Software Engineering Intern

June 2017 - September 2017

*Google**Mountain View, CA*

- Worked on Project Tango (later Daydream), researching machine learning techniques to improve inertial measurement unit (IMU) accuracy on smartphones.
- Implemented IMU data processing pipeline to analyze sensor performance across diverse real-world scenarios.

EARLY EXPERIENCE

Undergraduate Researcher

August 2013 - July 2014

*UCLA Vision Lab under Professor Soatto**Los Angeles, CA***Android Developer Lead**

October 2012 - October 2013

*UCLA Ozcan Lab**Los Angeles, CA***Software Engineering Intern**

June 2013 - August 2013

*Google**Venice Beach, CA***Software Engineering Intern**

June 2012 - August 2012

*Zynx Health**Los Angeles, CA***AWARDS AND HONORS**

Outstanding Reviewer of 3DV 2020**National Science Foundation Graduate Research Grant Honorable Mention****Outstanding Bachelor of Science Degree Award (Computer Science)****OUTREACH VOLUNTEERING**

Co-director of Penn Open Labs

October 2016 - January 2020

*University of Pennsylvania***Robot Design Judge for FIRST Lego League Regional Championship**

February 2014 - February 2019, three times

*University of Pennsylvania***Mentor for NSF Research Experience for Teachers (RET)**

July 2016 - August 2016

*University of Pennsylvania***CHARITY PROJECTS**

Project: The First Science of Savings Challenge

September 2020

*SaverLife***Project: 24 Hours of Good**

November 2013

*Google in Venice Beach, CA***Project: San Diego Zoo App**

March 2011- June 2011

University of California, Los Angeles